

Highlights

Literature-Grounded Research Ideation with State-of-the-Art Pipelines and Facet-Grounded Gap Verification

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- Explicit SOTA tier definition with dual-pass retrieval and hybrid re-ranking over open scholarly catalogs.
- Reproducible landscape $\rightarrow$ gap matrix $\rightarrow$ proposal pipeline with evidence-only generation rules.
- FGGV extension for facet-level novelty, gap alignment, and diversity-aware recommendation lists.
- Formal notation, equations, algorithms, and online evaluation protocol for transparent auditing.

Literature-Grounded Research Ideation with State-of-the-Art Pipelines and Facet-Grounded Gap Verification

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Abstract

Many computational techniques have been proposed to help researchers explore scientific directions, yet the volume of published literature often exceeds what any individual can survey before proposing new work. This problem, known as *literature overload*, makes it difficult to filter relevant papers, identify open gaps, and generate ideas that are genuinely grounded in evidence. This paper presents POST-REC (Paper-Oriented Scientific Topic Recommender), a reproducible ideation system that models research proposal generation as a recommender problem over literature-backed items. The proposed method combines dual-pass bibliographic retrieval over OpenAlex, deterministic relevance filtering, hybrid BM25–dense–vector re-ranking with explicit state-of-the-art (SOTA) tier quotas, and a multi-stage pipeline that materializes landscape summaries and gap matrices before idea generation. An extension called *Facet-Grounded Gap Verification* (FGGV) adds literature facet maps, gap–facet alignment, contrastive facet novelty indices, and diversity-aware selection. We formalize notation, component metrics, composite ranking equations, and pseudocode algorithms. A motivation scenario in computer science illustrates relevance scoring, gap-matrix construction, and facet-level verification. The deployed system evaluates methods online through a blind A/B study between SOTA and FGGV, collecting volunteer feedback via the Expectation Alignment Score and related metrics.

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1. Introduction

The exponential growth of scholarly publishing has increased the difficulty of identifying relevant literature, recognizing underserved research gaps, and proposing novel directions that remain faithful to current evidence [1, 2]. A computer-science researcher investigating retrieval-augmented generation (RAG) for literature-grounded research ideation, for example, may need to analyze thousands of papers on LLM agents, hybrid retrieval, and novelty verification before articulating a feasible proposal. Too much information can quickly exceed cognitive limits, while unconstrained large language model (LLM) outputs may invent citations, recycle saturated ideas, or hide how “state of the art” was defined [3].

Recommender systems (RSs) have emerged as a practical response to overload problems by filtering items according to user preferences [4, 1]. In POST-REC, the recommended *items* are structured research ideas; user preferences are expressed through seed topics, research area, depth, constraints, and subsequent feedback. Retrieval-augmented generation (RAG) [5] grounds generation in retrieved documents, but RAG alone does not specify SOTA tiers, explicit gap matrices, or facet-level verification. Recent ideation agents—including ResearchAgent [6], NOVA [7], SciMON [8], Scideator [9], and the Idea Novelty Checker [10]—improve retrieval and novelty assessment, yet rarely expose all methodological artifacts within one reproducible pipeline.

This paper aims to investigate, model, and quantify literature-grounded research ideation by proposing a transparent pipeline that:

1. defines SOTA through configurable recency and citation tiers;
2. constructs an explicit gap matrix before proposal generation;
3. verifies novelty at document and facet levels;
4. learns from volunteer feedback through an Expectation Alignment Score (EAS).

The main contributions of this work are:

- **The SOTA-grounded ideation pipeline**, which produces landscape summaries and a gap matrix before generating proposals linked to concrete evidence papers;

- **The hybrid retrieval and relevance model**, combining dual-pass OpenAlex fetching, lexical relevance scoring, BM25 + dense embedding + pgvector ranking, and SOTA tier quotas;
- **Facet-Grounded Gap Verification (FGGV)**, extending the baseline with literature facet maps (LFMs), gap–facet alignment (GFA), contrastive facet novelty (CFN), and facet diversity selection (FDS);
- **An online evaluation protocol**, centered on blind A/B assignment between SOTA and FGGV in Auto mode, with volunteer feedback aggregated on a validation dashboard.

The remaining aspects of this paper are structured as follows. Section 2 discusses related work. Section 3 introduces the proposed system, including notation, equations, algorithms, and a motivation scenario. Section 4 presents the experimental evaluation. Section 5 discusses limitations, and Section 6 concludes the paper and presents future work.

2. Related work

In the realm of literature-grounded ideation, numerous studies employ retrieval, planning, or facet-based verification to recommend research directions. However, a significant gap remains in jointly exposing explicit SOTA tiering, gap-matrix artifacts, and facet-level verification within one pipeline.

Table 1 compares representative systems. Si et al. [2] provide rigorous human evaluation but no facet map or gap matrix. ResearchAgent [6] and NOVA [7] emphasize iterative expansion over graphs or planned search at higher computational cost. Scideator [9] and the Idea Novelty Checker [10] apply facet-level checks, typically post-hoc rather than during gap-driven generation. SciMON [8] optimizes contrastive novelty against inspiration papers but does not materialize a landscape matrix artifact. Chain of Ideas [11] models temporal literature chains without an explicit gap cell structure.

Unlike these approaches, POST-REC materializes `sota_landscape` and `gap_matrix` JSON artifacts before proposal generation, making gap-driven ideation inspectable by reviewers. Hybrid lexical–neural retrieval follows established practice in scientific search [12], while facet re-ranking relates to RankGPT-style rerankers [13].

Table 1: Comparison of literature-grounded ideation approaches (2023–2026).

System	Retrieval	Gap structure	Novelty unit	Limitation
Si et al.	Semantic Scholar API	Implicit in abstracts	Human-rated originality	No facet map
ResearchAgent	Citation graph	Agent review loops	Multi-agent debate	High cost
NOVA	Planned embedding search	Iterative broadening	Planning steps	Weak SOTA tiering
Scideator	Paper-pool facets	UI recombination	Facet-level checks	Weak gap alignment
INC	Retrieve–rerank	Post-hoc	Per-facet LLM compare	Not gap-driven
POST-Rec SOTA	OpenAlex dual-pass + hybrid rank	Landscape + gap matrix	Document + verified signals	LLM-synthesized gaps
POST-Rec FGGV	Same as SOTA	Gap matrix + LFM	Per-facet FNI + GFA	Online sample size

3. The proposal

This section outlines the proposed POST-REC system. Firstly, the recommendation overview is presented. After that, the user expectation model and bibliographic data features are described. Following this, the discussion moves to document preprocessing, state-of-the-art tier definition, hybrid re-ranking, generation modes, landscape and gap-matrix construction, proposal generation, verified ranking, and the Facet-Grounded Gap Verification (FGGV) extension. Finally, a motivation scenario with numeric illustration and the recommendation algorithms are presented.

3.1. Recommendation overview

The proposed system treats research ideation as a recommendation problem: the items to be recommended are structured research ideas, and user

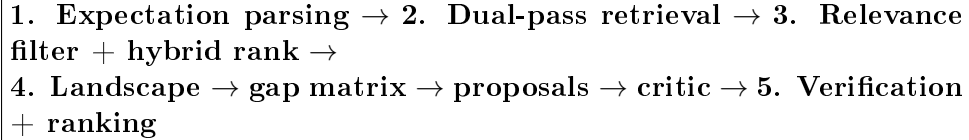


Figure 1: Overview of the POST-REC recommendation pipeline.

preferences are expressed through seed topics, research area, depth, constraints, and subsequent feedback. Given a user expectation E , the pipeline produces a ranked list of ideas linked to evidence papers retrieved from open scholarly catalogs.

Figure 1 summarizes the main stages. The deployed application reports progress to the user at each step.

The stages can be described as follows:

1. **Expectation parsing.** The system reads seed topics, research area, depth, and optional constraints. Topics learned from prior positive feedback are merged with the seed set (Eq. (2)).
2. **Dual-pass retrieval.** OpenAlex [14] is queried twice: once for recent state-of-the-art (SOTA) papers and once for seminal foundation papers. Results are merged and deduplicated.
3. **Relevance filtering and hybrid re-ranking.** Each paper receives a lexical relevance score. Papers below a fixed threshold are removed. The remaining set is re-ranked with BM25, dense embeddings, and vector search, while enforcing a SOTA tier quota in the context window.
4. **Mode-dependent generation.** Depending on the selected run mode, the system either performs a single-pass generation (Quick mode) or a multi-stage pipeline with landscape summary, gap matrix, and proposals (SOTA, Exploratory, and FGGV modes).
5. **Verification and ranking.** Each idea is checked against evidence rules, scored on multiple dimensions, and ranked. In FGGV mode, facet-level novelty and gap alignment are added before diversity-aware selection.

This design separates retrieval from generation and makes intermediate artifacts available for inspection. A reviewer can therefore examine which papers were retained, how gaps were identified, and why a given idea received its final score.

3.2. User expectation model

A researcher U is represented at run time by an expectation

$$E = \langle T_{\text{seed}}, A, O, D, C \rangle, \quad (1)$$

where T_{seed} is the set of seed topic phrases, A is the research area, O is the ideation objective (e.g., a new project direction or a literature survey), D is the desired depth, and C is a set of optional constraints (maximum papers, excluded venues, and similar limits).

After each run, the user may rate individual ideas. Ratings include usefulness, relevance, clarity, feasibility, trust, and a would-use indicator. These values are combined into the Expectation Alignment Score (EAS), defined in Section 4. Positive feedback updates T_{learned} ; negative feedback updates avoided topics. Both lists influence future retrieval through the query token set Q and the penalty term p_{avoid} in Eq. (5). In this way, the system learns from repeated use without requiring explicit profile editing.

3.3. Bibliographic data features

POST-REC uses OpenAlex as its primary literature source, with optional normalization from arXiv, Semantic Scholar, and Crossref metadata. Each document p is represented by features used across retrieval, ranking, and generation:

- **Text features:** title, abstract, and venue for lexical overlap and embedding;
- **Bibliographic features:** DOI, publication year, and citation count for deduplication and tier assignment;
- **Domain features:** field, subfield, and topic tags for alignment filtering.

Only papers that pass relevance filtering and hybrid ranking are passed to the generation stage. This rule reduces the risk of generating ideas that cite papers unrelated to the user’s topics.

Table 2: Notation used in the proposed method.

Symbol	Description
U	Researcher user
E	User expectation $\langle T_{\text{seed}}, A, O, D, C \rangle$
T_{seed}	Seed topic phrases
A	Research area
Q	Query token set for retrieval and relevance
$p \in \mathcal{D}$	Normalized paper
$O(X, Q)$	Token overlap ratio $ X \cap Q / Q $
$s_{\text{rel}}(p)$	Paper relevance score in $[0, 1]$
e_p	Embedding of paper p
$\mathcal{S}_{\text{SOTA}}$	SOTA-recent tier subset
$\mathcal{S}_{\text{found}}$	Foundation (seminal) tier subset
G	Gap matrix from landscape analysis
FNI_f	Facet Novelty Index for facet f
GFA	Gap-facet alignment score
S_{verified}	Verified final ranking score

3.4. Notation

Understanding the notation is crucial for comprehending the equations and algorithms presented below. Table 2 summarizes the main symbols.

The expanded topic set is defined as

$$T^* = \text{unique}(T_{\text{seed}} \cup T_{\text{learned}} \cup T_{\text{techniques}}), \quad (2)$$

and query tokens are $Q = \text{tokens}(T^* \cup A)$ after stopword removal and token-length filtering ($|\text{token}| > 2$). The retained paper set is

$$\mathcal{P} = \{p \in \mathcal{D} \mid s_{\text{rel}}(p) \geq \tau_{\text{rel}}\}, \quad (3)$$

with default threshold $\tau_{\text{rel}} = 0.22$.

3.5. Document preprocessing and relevance filtering

Retrieved papers are normalized (title, DOI, year, source), deduplicated, and scored before any language-model call. For each paper p , token sets

$T_{\text{title}}(p)$ and $T_{\text{body}}(p)$ are built from the title and from title||abstract||venue, respectively. The overlap function is

$$O(X, Q) = \frac{|X \cap Q|}{|Q|}, \quad Q \neq \emptyset. \quad (4)$$

The relevance score combines title overlap, body overlap, a citation boost, and an optional penalty for avoided topics:

$$s_{\text{rel}}(p) = w_t O(T_{\text{title}}, Q) + w_b O(T_{\text{body}}, Q) + b_{\text{cite}} - p_{\text{avoid}}, \quad (5)$$

$$b_{\text{cite}} = \min\left(\frac{\ln(1 + c_p)}{\delta}, b_{\text{max}}\right), \quad (6)$$

with defaults $w_t = 0.55$, $w_b = 0.30$, $\delta = 12$, $b_{\text{max}} = 0.15$, and $p_{\text{avoid}} = 0.35$ when the paper matches an avoided topic. If both title and body overlap are very low, the score is capped at 0.18, so that highly cited but off-topic papers are not retained.

This step is important because citation count alone is a weak signal of relevance. A seminal textbook on information retrieval, for example, may have thousands of citations yet contribute little to a run on retrieval-augmented research agents. The relevance filter therefore prioritizes lexical alignment with the user’s query before any generative step.

3.6. State-of-the-art tier definition

A central design choice in POST-REC is to define SOTA membership explicitly rather than to rely only on retrieval ranking. Two tiers are used:

$$p \in \mathcal{S}_{\text{SOTA}} \Leftrightarrow \text{year}(p) \geq y_{\text{now}} - Y, \quad (7)$$

$$p \in \mathcal{S}_{\text{found}} \Leftrightarrow c_p \geq \tau_c, \quad (8)$$

with defaults $Y = 3$ years and $\tau_c = 50$ citations. Papers in $\mathcal{S}_{\text{SOTA}}$ represent recent work; papers in $\mathcal{S}_{\text{found}}$ represent established references that provide context but may not reflect the latest methods.

Dual-pass retrieval queries OpenAlex separately for each tier and then merges the results. After hybrid ranking, a quota $\phi = 0.6$ ensures that at least 60% of the top- k context papers belong to $\mathcal{S}_{\text{SOTA}}$. Each generated idea must also link at least one SOTA anchor from the retrieved set. These rules make the notion of “current work” inspectable and reproducible.

Table 3: Generation modes in POST-REC.

Mode	Pipeline	Gap matrix	Critic	Ranking focus
Quick	Single-pass LLM	No	Standard	Relevance
SOTA	Landscape → gaps → proposals	Yes	Strict	SOTA fit
Exploratory	Same as SOTA	Yes	Relaxed	Novelty
FGGV	SOTA + facet verification	Yes	Strict	FGGV score

3.7. Hybrid re-ranking

Each retained paper $p \in \mathcal{P}$ receives a sparse score $s_{\text{BM25}}(p, Q)$ and a dense score $s_{\text{dense}}(p, Q) = \cos(e_p, e_Q)$, where e_p and e_Q are embedding vectors. The hybrid score blends both signals (sparse weight 0.4 by default) with pgvector retrieval (blend factor 0.25). This approach follows established practice in scientific search, where lexical and neural signals capture complementary aspects of relevance [12].

After sorting, the tier quota is applied so that the generation context is not dominated by older seminal papers. The resulting ranked set $\mathcal{P}_k$ is the input to all multi-stage generation modes.

3.8. Generation modes

POST-REC supports four generation modes. Table 3 summarizes their main differences. All modes share the same retrieval and relevance stages described above.

Quick mode performs a single generative pass over the top-ranked papers. It is faster and suitable for an initial preview of possible directions. It still applies evidence-only rules and novelty verification, but it does not build a landscape or gap matrix.

SOTA mode runs the full multi-stage pipeline: landscape summary, gap matrix, proposal generation, and critic filtering. The critic operates in strict mode, meaning that ideas with validation issues are marked as needing refinement rather than being silently dropped.

Exploratory mode uses the same pipeline as SOTA but assigns higher weight to novelty dimensions in the final ranking and applies a relaxed critic.

This mode is intended for users who prefer broader ideas and accept that more proposals may require manual editing.

FGGV mode extends the SOTA pipeline with facet-level verification, as described in Section 3.12. It is the proposed method evaluated against SOTA in the online study.

When the user selects **Auto mode**, the system does not choose among all four pipelines. Instead, it assigns the run to SOTA (control) or FGGV (treatment) through a blind experiment, as described in Section 4. Quick and Exploratory remain available as explicit user choices outside the experiment.

3.9. Landscape and gap matrix construction

Let $\mathcal{P}_k$ denote the top- k papers after hybrid ranking. In SOTA, Exploratory, and FGGV modes, the system performs three preparatory steps before generating proposals.

First, papers in $\mathcal{P}_k$ are grouped into thematic clusters. For each cluster, the system summarizes the main problems addressed, methods used, datasets, evaluation protocols, and reported limitations. This summary is stored as a `sota_landscape` artifact.

Second, the system builds a gap matrix G with rows corresponding to sub-problems and columns corresponding to method, data, and evaluation families. Each cell G_{ij} is labeled as *well-covered*, *partial*, or *underserved*, depending on how densely the retrieved literature addresses that combination. The matrix is stored as a `gap_matrix` artifact.

Third, the generator is instructed to sample underserved cells when drafting proposals. Each proposal must declare how it differs from prior work (`novelty_delta`), which gaps it addresses (`aligned_gaps`), and which retrieved papers support it (`sota_anchors` and `evidence_papers`). These fields make the reasoning behind each idea visible to the user and to external reviewers.

Unlike systems that embed gap information only inside a prompt, POST-REC stores the landscape and gap matrix as structured outputs. This design supports auditing and allows downstream components (such as GFA in FGGV mode) to compare proposals against the same gap structure.

3.10. Proposal generation and critic

Proposal generation is performed by a language model conditioned on E , $\mathcal{P}_k$, and G . The model must obey evidence-only rules: every cited paper must

appear in $\mathcal{P}_k$, and invented DOIs or authors are rejected by deterministic validation.

After initial generation, a critic pass reviews each proposal. The critic checks grounding in the retrieved set, overlap with existing papers, and completeness of required fields. In strict modes (SOTA and FGGV), proposals that fail validation receive publication status `needs_refinement`. In Exploratory mode, only citation errors trigger refinement; other issues are retained for user review.

Novelty is also assessed at document level. If the embedding of a proposal is too close to a single retrieved paper, the idea may be down-ranked or rejected in SOTA mode. This step reduces near-duplicate suggestions that rephrase one source paper.

3.11. Verified ranking

Each idea i receives language-model estimates for several dimensions $\text{dim}_{i,j} \in [0, 100]$, including relevance, novelty, evidence, feasibility, trend, publication potential, and strategic fit. Two additional verified signals are computed: `SOTA_fit( $i$ )`, which measures alignment with recent anchors, and `novelty_verified( $i$ )`, which combines embedding distance and claim overlap checks. The verified score is

$$S_{\text{verified}}(i) = \sum_j w_j \cdot \text{dim}_{i,j} + w_{\text{sota}} \cdot \text{SOTA_fit}(i) + w_{\text{nov}} \cdot \text{novelty_verified}(i). \quad (9)$$

The weights w_j , w_{sota} , and w_{nov} depend on the run mode. SOTA mode assigns higher weight to `SOTA_fit` and `novelty_verified`; Exploratory mode increases the novelty weights; Quick mode uses a balanced default profile.

The verified score is the baseline ranking function for Quick, SOTA, and Exploratory modes. FGGV mode uses Eq. (9) as one component of a larger composite, as described next.

3.12. Facet-Grounded Gap Verification

Facet-Grounded Gap Verification (FGGV) extends the SOTA pipeline with four components that address limitations observed in recent ideation systems [2, 9, 10]. Existing methods often check novelty at document level or apply facet comparisons only after generation. FGGV integrates facet structure into both generation and ranking.

3.12.1. Literature Facet Map (LFM)

For each paper in $\mathcal{P}_k$, the system extracts four facets: problem, method, data, and evaluation. These facets are indexed at corpus level to form a Literature Facet Map (LFM). Each facet entry stores the textual statement and a saturation score that reflects how often similar statements appear in the retrieved set. Saturation helps identify crowded research directions (many similar method statements) and sparse directions (few statements, potential gaps).

The LFM is built once per run and reused by all FGGV scoring components. It provides a structured view of what the literature already claims along each facet dimension.

3.12.2. Gap-Facet Alignment (GFA)

Each proposal declares `facet_deltas` (what changes along each facet) and `aligned_gaps` (which gap-matrix cells the idea targets). The Gap-Facet Alignment score (GFA) measures token overlap between the proposal facets and the gap statements extracted from G . When a proposal explicitly lists aligned gaps, the alignment score receives full weight; otherwise it is reduced. GFA therefore rewards ideas that address gaps identified in the landscape stage, not only ideas that appear novel in isolation.

3.12.3. Contrastive Facet Novelty (CFN)

Document-level novelty can be high even when individual facets repeat well-known claims. Contrastive Facet Novelty addresses this by computing a Facet Novelty Index (FNI) per facet. For facet f ,

$$\text{FNI}_f(i) = 1 - \max_{s \in \text{LFM}_f} \text{sim}(\text{facet}_f(i), s), \quad (10)$$

where `sim` is cosine similarity when embeddings are available, or token Jaccard similarity otherwise. The overall FNI is a weighted sum across facets, with method (0.35) and problem (0.30) receiving the highest weights, followed by evaluation (0.20) and data (0.15).

Saturation-aware weighting adjusts FNI toward under-served facets: novelty in a sparse facet is boosted, while novelty in a crowded facet is penalized. An optional language-model facet critic provides an additional per-facet judgment, blended at 25% with the contrastive score. If two or more facets fall below a minimum novelty threshold, the proposal is flagged for refinement.

3.12.4. Facet Diversity Selection (FDS)

After scoring, FGGV mode may produce more candidates than the user requests. Facet Diversity Selection applies maximal marginal relevance (MMR) on facet token sets to choose a diverse top- k list. This step addresses a common limitation in ideation systems, where several highly ranked ideas differ only slightly in method or problem framing [2].

3.12.5. Composite FGGV score

The final FGGV score combines verified ranking, facet novelty, gap alignment, and document novelty:

$$\text{FGGV}(i) = \alpha S_{\text{verified}}(i) + \beta \text{FNI}(i) + \gamma \text{GFA}(i) + \eta \cdot \text{doc_novelty}(i), \quad (11)$$

with defaults $\alpha = 0.30$, $\beta = 0.32$, $\gamma = 0.23$, and $\eta = 0.15$, where S_{verified} is normalized to $[0, 1]$ before weighting and the final score is scaled to $[0, 100]$. All sub-scores are stored in the run output so that reviewers can inspect how each term contributed to the final ranking.

3.13. Feedback and profile learning

After a run completes, the user may rate each idea. The Expectation Alignment Score (EAS) combines six ratings into a single value, as defined in Eq. (12) and Algorithm 3. Ideas marked as useful contribute terms to T_{learned} ; ideas marked as not useful contribute to avoided topics. On the next run, learned topics expand the query set in Eq. (2), while avoided topics reduce relevance through p_{avoid} in Eq. (5). This closed loop allows the system to adapt to individual researchers without manual profile maintenance.

3.14. Motivation scenario

To demonstrate how the proposed method operates in practice, this subsection walks through a complete run in computer science. The example covers relevance filtering, tier selection, gap-matrix construction, and facet scoring under FGGV mode.

Consider a researcher U who submits the following expectation:

- $T_{\text{seed}} = \{\text{retrieval-augmented generation, research ideation agents}\};$
- $A = \text{artificial intelligence};$
- $\text{mode} = \text{FGGV}.$

Step 1: Topic expansion. Suppose the user has no prior feedback. Then $T^* = T_{\text{seed}}$ and the query token set Q is built from T^* and A after stopword removal.

Step 2: Relevance filtering. Consider two retrieved papers.

Paper p_1 (retained): title “*Benchmarking retrieval-augmented generation pipelines for scientific question answering*”, year 2024, citations 92. Token overlap yields $O(T_{\text{title}}, Q) = 0.45$ and $O(T_{\text{body}}, Q) = 0.31$. With $b_{\text{cite}} = 0.15$,

$$s_{\text{rel}}(p_1) = 0.55(0.45) + 0.30(0.31) + 0.15 = 0.490 > \tau_{\text{rel}}.$$

Paper p_2 (filtered): title “*Introduction to classical information retrieval models*”, year 1999, citations 680. Although $b_{\text{cite}} = 0.15$, weak overlap caps $s_{\text{rel}}(p_2) \leq 0.18$. Paper p_2 is excluded despite its high citation count.

Step 3: Hybrid ranking and tier quota. After filtering, suppose $|\mathcal{P}| = 52$, with 31 recent SOTA papers and 21 seminal papers. With context size $k = 30$ and quota $\phi = 0.6$, at least 18 recent SOTA papers enter the generation context.

Step 4: Gap matrix. Table 4 shows an excerpt of the gap matrix G produced for this run. The cell at (literature-grounded ideation, facet verification) is labeled *underserved*, indicating that few retrieved papers combine these dimensions.

Table 4: Excerpt of gap matrix G for the motivation scenario.

	Hybrid retrieval	Facet verification	Human eval.
Literature-grounded ideation	partial	underserved	partial
Scientific QA with RAG	well-covered	partial	underserved

Step 5: Proposal and FGGV scoring. Suppose the generator produces an idea i titled “*Facet-level gap verification for literature-grounded research agents*”, with `aligned_gaps` pointing to the underserved cell in Table 4. The method facet proposes multi-stage facet verification instead of single-pass RAG generation.

For this idea, $\text{GFA}(i)$ is high because the facet deltas overlap with the gap statement for literature-grounded ideation $\times$ facet verification. In the LFM, most method statements describe single-pass RAG pipelines; the proposal’s method facet is distant from these statements, yielding $\text{FNI}_{\text{method}}(i) = 0.76$. Assuming $S_{\text{verified}}(i) = 72$ (on a 0–100 scale), $\text{GFA}(i) = 0.85$, and

Algorithm 1 Paper relevance filtering

Require: Papers $\mathcal{D}$, query tokens Q , threshold τ_{rel}

Ensure: Filtered set $\mathcal{P}$

```
1:  $\mathcal{P} \leftarrow \emptyset$ 
2: for each  $p \in \mathcal{D}$  do
3:   Compute  $O(T_{\text{title}}, Q)$  and  $O(T_{\text{body}}, Q)$  using Eq. (4)
4:    $b_{\text{cite}} \leftarrow \text{Eq. (6)}$ 
5:    $s_{\text{rel}}(p) \leftarrow \text{Eq. (5)}$ ; apply weak-overlap cap
6:   if  $s_{\text{rel}}(p) \geq \tau_{\text{rel}}$  then
7:      $\mathcal{P} \leftarrow \mathcal{P} \cup \{p\}$ 
8:   end if
9: end for
10: return  $\mathcal{P}$ 
```

$\text{doc_novelty}(i) = 0.68$, the composite score from Eq. (11) is computed by normalizing the verified score to $[0, 1]$ and scaling the result back to $[0, 100]$:

$$\text{FGGV}(i) = 100 \times (0.30 \times 0.72 + 0.32 \times 0.76 + 0.23 \times 0.85 + 0.15 \times 0.68) \approx 75.7.$$

This idea would rank above a proposal that repeats a well-covered RAG-for-QA direction with lower FNI and GFA, even if both ideas appear novel at document level.

This example illustrates how lexical relevance, explicit tier quotas, gap-matrix alignment, and facet-level novelty interact in a single run. The same run would produce lower scores for ideas that cite valid papers but do not address identified gaps or that repeat saturated method statements from the LFM.

3.15. Recommendation algorithms

The main procedures of the proposed system are summarized in Algorithms 1–3. Algorithm 1 filters papers by relevance. Algorithm 2 performs multi-stage generation and mode-dependent scoring. Algorithm 3 updates the user profile from feedback.

4. Experimental evaluation

This section presents the evaluation of the proposed system. The principal validation path is **online**: researchers use the deployed platform, rate

Algorithm 2 SOTA-grounded idea generation

Require: Expectation E , ranked papers $\mathcal{P}_k$, mode m

Ensure: Ranked idea set $\mathcal{I}$

- 1: Build landscape summary from $\mathcal{P}_k$
 - 2: Construct gap matrix G
 - 3: Sample underserved cells from G
 - 4: $\mathcal{I}_0 \leftarrow \text{LLM-GENERATE}(E, \mathcal{P}_k, G)$ under evidence-only rules
 - 5: $\mathcal{I}_1 \leftarrow \text{CRITIC}(\mathcal{I}_0, \mathcal{P}_k)$
 - 6: **if** $m = \text{fggv}$ **then**
 - 7: Build LFM; compute FNI and GFA; apply FDS
 - 8: Score ideas with Eq. (11)
 - 9: **else**
 - 10: Score ideas with Eq. (9)
 - 11: **end if**
 - 12: **return** $\mathcal{I}$ sorted by final score
-

Algorithm 3 Expectation Alignment Score and profile update

Require: Ratings: usefulness U , relevance R , clarity C , feasibility F , trust T , would-use W

Ensure: EAS and updated topic profile

- 1: Map $W \in \{\text{yes, maybe, no}\}$ to $\{5, 3, 1\}$
 - 2: $\text{EAS} \leftarrow 0.25U + 0.20R + 0.20C + 0.15F + 0.10T + 0.10W$
 - 3: Update T_{learned} or avoided topics
 - 4: **return** EAS
-

generated ideas, and contribute feedback that is aggregated by experimental variant. Supplementary offline scripts support regression testing during development but are not the main source of evidence for method comparison.

4.1. Research questions

The evaluation is organized around four research questions:

- **RQ1:** Does FGGV improve user-rated alignment and originality compared with the SOTA pipeline?
- **RQ2:** Does facet-level verification reduce false-novel proposals compared with document-level novelty alone?

Table 5: Baseline methods for comparison.

ID	Mode	Description
B1	Quick	Single-pass RAG with document novelty verification
B2	SOTA	Landscape $\rightarrow$ gap matrix $\rightarrow$ proposals (control)
B3	—	RAG without verification (offline ablation only)
M	FGGV	SOTA pipeline plus facet verification (treatment)
M-GFA	FGGV	Ablation without gap-facet alignment
M-CFN	FGGV	Ablation without contrastive facet novelty
M-SAT	FGGV	Ablation without saturation-aware weighting

- **RQ3:** Does facet diversity selection improve the variety of ideas in the final list without reducing relevance?
- **RQ4:** Do FNI and GFA correlate with user-reported originality scores?

RQ1 is addressed directly by the online A/B study. RQ2–RQ4 are supported by stored FGGV sub-scores, offline ablations, and planned expert studies.

4.2. Evaluation design

Two evaluation channels are used. The *online channel* collects feedback from real users during normal system use. The *offline channel* replays fixed topic sets through Quick, SOTA, and FGGV pipelines to verify implementation consistency after code changes. Table 5 lists the baseline methods.

The online study compares B2 (control) and M (treatment). Offline ablations compare M with M-GFA, M-CFN, and M-SAT to isolate the contribution of each FGGV component.

4.3. Online evaluation setup

When a user selects **Auto** mode, the system assigns the run to one of two variants through a blind experiment:

- **Control (SOTA):** the full landscape $\rightarrow$ gap matrix $\rightarrow$ proposals $\rightarrow$ critic pipeline;
- **Treatment (FGGV):** the same pipeline plus LFM, GFA, CFN, and FDS.

Assignment uses a deterministic hash of the user identifier and the experiment identifier `fggv_vs_sota_v1`, with a default treatment fraction of 0.5. The same user always receives the same variant across runs (sticky assignment). Method labels, FGGV sub-scores, and run cost are hidden during blind runs so that user ratings are not biased by knowing which pipeline produced the ideas.

Quick and Exploratory modes are not included in Auto assignment. Quick differs structurally from the SOTA pipeline (single-pass versus multi-stage), and Exploratory differs only in ranking weights and critic strictness. Including them in random assignment would mix several effects and make the comparison between SOTA and FGGV harder to interpret. Users who want those modes select them explicitly; such runs are excluded from the A/B analysis.

4.4. Evaluation metrics

The primary online metrics are as follows.

Expectation Alignment Score (EAS). For each rated idea, EAS combines six user judgments:

$$\text{EAS} = 0.25U + 0.20R + 0.20C + 0.15F + 0.10T + 0.10W, \quad (12)$$

where U is usefulness, R is relevance, C is clarity, F is feasibility, T is trust, and W is the would-use indicator (mapped from yes/maybe/no to 5/3/1). EAS measures how well an idea matches the user’s stated expectation.

Originality. Users report how novel they consider each idea on a standard rating scale. This metric addresses RQ1 and RQ4 directly.

Approval rate. The proportion of ideas marked as suitable for use in a paper or project. This metric captures practical utility beyond alignment.

Secondary metrics stored for FGGV runs include per-facet FNI, GFA, anchor validity (share of ideas with verifiable SOTA anchors), and intra-list facet diversity (average pairwise Jaccard distance). These metrics support RQ2–RQ4 and offline ablation analysis.

Feedback updates T_{learned} and avoided topics, which influence subsequent retrieval through Eq. (2) and Eq. (5). The validation dashboard aggregates mean EAS, originality, and approval rate by `experiment_variant`.

4.5. Results and monitoring

In the current pilot deployment, run counts and mean metrics are monitored separately for the control (SOTA) and treatment (FGGV) variants. As the number of rated ideas grows, the system reports whether differences in EAS and originality are statistically significant. This continuous monitoring supports evaluation under realistic usage conditions rather than relying solely on a fixed offline benchmark with synthetic judges.

Preliminary observations from the pilot indicate that both variants produce evidence-backed ideas, but FGGV runs tend to surface proposals with more diverse facet profiles in the final list. Formal superiority claims require a larger sample of blind ratings; until then, the online study should be interpreted as an ongoing data collection effort.

4.6. Supplementary offline evaluation

The repository provides an offline harness (`scripts/run_offline_evaluation.py`) that replays golden topics through Quick (B1), SOTA (B2), and FGGV (M), including ablation variants. This harness verifies that scoring and retrieval changes do not break expected ordering on fixed fixtures. It also computes discriminability between strong and weak proposals on held-out examples.

Offline evaluation does not replace volunteer judgments. Its role is to catch regressions during development and to support ablation analysis (Table 5) while the online cohort accumulates blind ratings.

5. Discussion

This section discusses the main implications of the proposed methodology, its relation to prior work, and current limitations.

5.1. Interpretation of the proposal

The proposed system addresses literature-grounded ideation through three layers that build on one another. The first layer is retrieval: dual-pass fetching, relevance filtering, and hybrid re-ranking ensure that generation is conditioned on papers aligned with the user’s topics. The second layer is gap-driven generation: the landscape and gap matrix make the structure of

the retrieved literature explicit before any idea is drafted. The third layer, FGGV, adds facet-level verification so that novelty and gap alignment are assessed along problem, method, data, and evaluation dimensions rather than only at document level.

The motivation scenario in Section 3.14 shows that these layers interact in a concrete way. A paper with many citations is not retained if it lacks lexical overlap with the query. A proposal that appears novel as a whole may still rank lower if its method facet repeats saturated claims from the LFM. Conversely, a proposal that targets an underserved gap cell receives a higher GFA score even when its overall wording is similar to existing work.

This layered design differs from agents that perform retrieval and generation in one pass [2] and from systems that verify facets only after the user has already selected an idea [10]. By storing intermediate artifacts and scoring components, POST-REC supports inspection at each stage.

5.2. *Relation to prior work*

Compared with ResearchAgent [6] and NOVA [7], the proposed system places less emphasis on iterative graph expansion and more on explicit SOTA tiering and gap-matrix structure. Compared with Scideator [9], FGGV integrates facet maps with gap alignment rather than relying on facet recombination alone. Compared with SciMON [8], contrastive novelty operates per facet against a literature map rather than only against a set of inspiration papers.

The online A/B design follows common practice in recommender system evaluation: a control pipeline that represents the current best method (SOTA) is compared with a treatment that adds a specific innovation (FGGV). Auto mode implements this design without requiring users to understand the underlying assignment.

5.3. *Limitations*

Several limitations should be noted. First, retrieval quality depends on OpenAlex metadata completeness; missing abstracts or incorrect year values can affect tier assignment and relevance scores. Second, the gap matrix is synthesized by a language model from retrieved papers; it may omit gaps that exist outside the retrieved set or mislabel cell coverage. Third, facet extraction from papers and proposals is approximate; errors in facet boundaries propagate to FNI and GFA. Fourth, the online sample is still growing; self-selected topics and volunteer availability may introduce sampling bias.

Fifth, Quick and Exploratory modes are not part of the blind experiment, so the study does not directly compare all four generation modes under the same protocol.

Despite these limitations, the explicit notation, stored artifacts, and component scores provide a foundation for reproducible comparison as larger samples and expert studies become available.

6. Conclusion

This paper presented POST-REC, a literature-grounded research ideation methodology that models proposal generation as a recommender problem over evidence-backed ideas. The system combines dual-pass bibliographic retrieval, deterministic relevance filtering, hybrid re-ranking with explicit SOTA tier quotas, and a multi-stage pipeline that produces landscape summaries and gap matrices before generating proposals. The Facet-Grounded Gap Verification (FGGV) extension adds literature facet maps, gap-facet alignment, contrastive facet novelty indices, and diversity-aware selection to address limitations of document-level novelty assessment.

The proposal was described through user and paper models, component metrics, composite ranking equations, generation modes, pseudocode algorithms, and a worked scenario in computer science. The deployed system evaluates methods online through a blind A/B study between SOTA (control) and FGGV (treatment) in Auto mode, collecting volunteer feedback via the Expectation Alignment Score and related metrics. Supplementary offline harnesses support regression testing and ablation analysis during development.

Future work includes scaling the online evaluation cohort, conducting blind expert studies on matched topics, integrating citation-graph features for retrieval expansion, and comparing directly against external systems such as ResearchAgent and Scideator on shared topic sets. Multilingual catalog support and calibration of ranking weights from logged feedback are also planned. The open implementation aims to support reproducible comparison as the field of literature-grounded ideation continues to develop.

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